

Supplementary file S1

Cross-source battery cycle-life prediction with landmarked survival models

This file accompanies AE_paper_elsarticle_v4.tex. All analyses are retrospective and exploratory. The source-level integrated IPCW Brier score uses each source's own supported horizon grid; it is not a common cross-source integral.

S1. Source characteristics and landmark eligibility

Source	Verified cells	Cycle-100 eligible/events	Recorded chemistry	Temperature (C)	Group or protocol labels	Median record density
MIT	124	124/43	LFP	30	one selected group (b)	1.000
XJTU	55	39/21	NCM532	unavailable	six batch labels	1.000
TJU	130	119/75	NCA, NCM, mixed NCM/NCA	25, 35, 45	three subsets; chemistry and protocol confounded	1.000
NASA	34 unique	8/3	cathode unspecified in certified local metadata	4, 24, 43, 44	six archive groups	1.000
CALCE	9	9/9	LCO (CS/CX), LFP (K2)	unavailable (RT)	one RT label across subsets	0.990
HUST	77	77/77	LFP	30	heterogeneous multistage discharge; no usable split	1.000

The observation-density column is the median ratio of verified unique cycle records to the integer-cycle span from first to last record across unique cells in a source. It describes the processed record grid, not original instrument sampling frequency. Empty temperature entries mean the verified local table did not certify a value.

At cycle 100, NASA has 34 unique cells: 8 eligible, 11 with the three-observed-record EOL at or before the landmark, and 15 without follow-up. XJTU excludes 16 cells with too few early observations; TJU excludes 11 without post-landmark follow-up. Among 376 eligible cells, 228 have observed EOL and 148 are right-censored. The main-text cohort figure is figures_v4/fig_v4_source_cohort.pdf.

S2. Benchmark and paired controls

benchmark_metrics.csv contains cycle-50/100 five-model scores, Harrell C, event-only MedAPE and supported-horizon counts. benchmark_predictions.csv gives one prediction per eligible target cell and model. NASA has only the cycle-125 supported horizon, so integrated Brier is not estimable; its Harrell C and MedAPE are descriptive values from eight cells and three events.

evidence08_metrics.csv, evidence08_predictions.csv, and evidence08_sensitivity.csv give within-source AFT and event-only regression comparisons. The within-source arm uses labels from other target cells and a different training size from the zero-shot arm. Event-only regression borrows the source-selected AFT scale to form a lognormal probability proxy; its Brier value is conditional on that choice. Ten fold partitions and twenty paired seeds reuse the same cells and do not increase independent sample size. XJTU/TJU protocol-group results and group counts are in evidence08p_groups.csv; TJU group membership is also a chemistry distinction.

S3. Endpoint, source correction and risk calibration

endpoint_common_cohort.csv identifies the common cohort across one-, three-, and five-observed-record endpoint definitions. endpoint_model_metrics.csv reports fixed-parameter Cox and XGBoost AFT refits on common cells and within-source common scoring horizons. For XJTU Cox at cycle 100, the three-record and five-record scores are 0.0823095 and 0.0708729 (displayed as 0.082 and 0.071). Different endpoint rules define different outcomes; a lower value does not validate one rule as more physical. The strict adjacent-integer-cycle rule has not been refitted.

source_correction_metrics.csv and source_correction_cohort.csv compare verified and historical source training on identical verified test labels and cells. The cycle-100 XGBoost AFT score improves on HUST (0.132 to 0.117) and worsens on CALCE (0.194 to 0.254). benchmark_horizons.csv gives cycle-500 CALCE predicted/held-out KM risk 0.672/0.111 and cycle-2000 HUST 0.451/0.896. These are fold-level risks, not individual calibration curves.

S4. Target-label adaptation and conformal count

recalibration_scores.csv, recalibration_horizons.csv, seed_target_recalibration_scores.csv, and split-membership outputs preserve each draw and paired score. Calibration draws are random and unstratified. At the cycle-100 20%

budget, calibration counts are MIT 25, XJTU 8, TJU 24, NASA 2, CALCE 2, and HUST 16. Across 20 training seeds and 30 overlapping draws, mean shifted-minus-baseline integrated Brier differences are +0.0031, +0.0076, +0.0040, unavailable, -0.0757, and +0.0079 in that order. These draws are not 600 independent experiments per source.

calibration_budget_feasibility.csv and protocol_v4.json document the separate event-only absolute-residual finite-quantile count calculation. A conventional empirical 95% split-conformal quantile requires at least 19 calibration event residuals. Under random 20% target-cell allocation, the chance of reaching that count is 5.22% for TJU. This does not estimate or guarantee coverage for censored target cells.

S5. Data access and declarations

The source repository and paper references appear in the main text. v4_artifact_manifest.csv records hashes for the internal analysis package. Source-specific redistribution permissions for MIT, NASA and CALCE, and the experiment-specific CALCE publication, remain under review. A public code/data repository and DOI have not yet been assigned. Real author names, contributions, funding and competing-interest statements require author confirmation before submission.

S6. Graphical abstract

Use figures_v4/graphical_abstract_v4.pdf or .png if a graphical abstract is submitted. The v3 graphical abstract refers to an older analysis and must not accompany this manuscript.